Static Voltage Stability Analysis of Large Scale Renewable Energy delivered via multi-voltage-level transmission sections

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***Abstract*—This paper addresses the static voltage stability constraints and complex coupling relationships in multi-voltage-level transmission sections for large-scale renewable energy integration systems. The study extracts two typical topological structures—parallel coupling and series coupling sections—from actual grid features and analyzes their interactive mechanisms affecting static voltage stability limits. Research reveals that in parallel systems, power flow distribution significantly impacts the overall transmission limit, with optimal limits achieved when heavily loaded branches concentrate on low-impedance paths. In series systems, the contribution of proximal station power growth to total transmission capacity exhibits nonlinear variation (sensitivity decreases with increasing output). Validated through actual grid simulations, the sensitivity of key node power flow to section limits is quantified, demonstrating that uniform power verification may cause conservatism or security risks. The findings provide theoretical support for online assessment of multi-level transmission section limit and active power optimization control, enhancing utilization rates while ensuring security to promote renewable energy accommodation.**

Keywords—renewable energy, static voltage stability, multi-voltage-level transmission sections

I. INTRODUCTION

Due to the uneven distribution of resources, renewable energy bases are often geographically distant from load centers in many cases, making direct power supply to end-users challenging. Consequently, it is necessary to connect energy bases with load centers through multi-voltage-level high-voltage transmission lines and substations to achieve long-distance power transmission and dispatch. With the increasing scale of renewable energy installations and the growing hierarchy of power aggregation, the scenario of delivering renewable energy through multi-voltage-level transmission sections has become increasingly prevalent [1-2]. In such delivery systems, voltage security issues emerge as critical constraints limiting the power transfer capacity of transmission sections at all levels, primarily due to the absence of conventional power support and the extended electrical distance of renewable energy delivery corridors [3].

In practical operations, voltage security verification is predominantly incorporated into renewable energy operational limit assessments through offline calculations. Among these, static voltage stability constraints constitute one of the primary operational limitations for renewable energy-integrated grids [4]. Taking China's Jibei regional power grids as an example, static voltage stability constraints serve as decisive factors in determining power transfer limits for certain renewable energy delivery sections. Analytical methods for static voltage stability problems mainly include the continuation power flow method [5], Jacobian matrix singularity method [6], and sensitivity analysis method [7]. The PV curve provides an intuitive understanding of static voltage stability by illustrating how system voltage varies and potentially collapses as wind power active output increases. Ref [8] demonstrates that as system operation approaches static voltage stability limits, voltage fluctuations caused by equivalent wind power active power variations become significantly amplified.

Static voltage stability exhibits strong coupling with power flow distribution across transmission sections. This results in varying static voltage stability characteristics among different voltage-level transmission sections within the same region as their power flow distributions change. Existing voltage security risk assessment methods, which assume uniform renewable energy generation across all sections, can only evaluate static voltage security risks under typical operating conditions. Such approaches may yield overly conservative limits that underutilize transmission capacity or produce optimistic estimates that jeopardize system operational security [9].

This paper conducts the analysis of static voltage stability in large-scale renewable energy delivery systems through multi-level transmission sections. First, by examining the topological characteristics of multi-voltage-level transmission networks, typical section coupling types are identified, with mechanistic analysis performed on the coupling relationships of transmission limits for each type. Case studies based on actual grid simulations are developed, employing sensitivity analysis to quantify the contributions of key node power flow states to section transfer capacity.

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II. THE STATIC VOLTAGE STABILITY LIMIT OF THE RENEWABLE ENERGY TRANSMISSION SYSTEM

The P-V curve method, as an effective approach for studying voltage stability issues in power systems, can indicate the critical voltage and limit power of load nodes, and provide corresponding voltage margin and active power margin, thus being widely applied. It establishes a relationship curve between the node voltage and the power flow of a regional load or transmission channel, thereby indicating the degree to which the regional load level or transmission channel power level causes the entire system to approach voltage collapse.

When applying the P-V curve method to analyze the static voltage stability of power grids with renewable energy integration, since the impact of renewable energy power injection on voltage stability needs to be considered, P represents the active power generated by new energy sources, while V can refer to the terminal voltage of generators, the point-of-common-coupling (PCC) voltage, or the voltage at other critical nodes in the collection system.

Due to the low-voltage ride-through (LVRT) strategies adopted by renewable energy systems, their transmission systems do not experience voltage instability in the traditional sense [8]. However, when the system approaches the inflection point of its P-V curve, active power fluctuations will cause significant voltage variations, which are detrimental to system operation. As shown in Fig.1, for a renewable energy transmission system in a certain region, when the transmitted power approaches the static voltage stability limit, the amplitude of voltage fluctuations caused by the same active power change increases significantly. Therefore, in actual operation, a margin is typically applied to the theoretical inflection point of the P-V curve to determine the practical static voltage stability limit for new energy transmission, as illustrated in Fig.2.

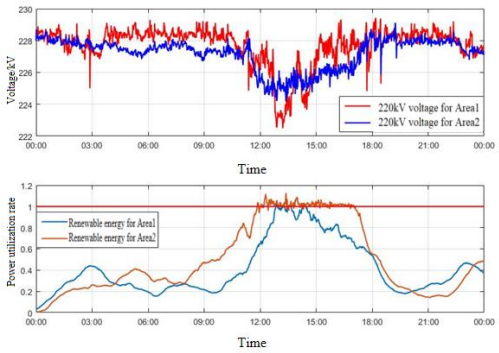


Fig. 1. Voltage fluctuation when the actual power aproaches the static voltage stability limit

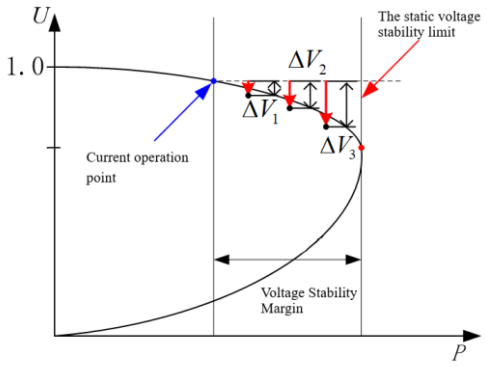


Fig. 2. Schematic diagram of P-V curve for renewable energy power transmission system

III. TOPOLOGICAL CHARACTERISTICS AND STATIC VOLTAGE STABILITY CALCULATION RESULTS OF LARGE-SCALE RENEWABLE ENERGY TRANSMISSION THROUGH MULTI-VOLTAGE LEVEL GRIDS

A. Analysis of Network Topological Characteristics

In a certain region, the main grid-connected voltage levels of new energy stations are 35kV, 110kV, and 220kV. The new energy power transmission sections are generally 220kV and 500kV, which are finally connected to the main grid through 1000kV tie lines. As low-voltage lines integrate into high-voltage lines, the network coupling relationships of multi-level sections become increasingly complex. As shown in Fig.3, a set is used to characterize the topological characteristics of multi-level coupling sections:

$$\begin{aligned} f(a) &= f(b1) * f(b2) \\ &= (\{C1, C2\} * f(c1)) * (\{F1, F2, F3\} * f(c2)) \\ &= (\{C1, C2\} * (\{A1, A2, A3\} * \{B1, B2\})) \\ &\quad * (\{F1, F2, F3\} * (\{D1, D2\} * \{E1, E2\})) \end{aligned} \tag{1}$$

The stations in access section a are described as the sets shown as $f(b1)$ and $f(b2)$. $f(b1)$ is further described as station $\{C1, C2\}$ and the set shown as $f(c1)$, while $f(b2)$ is further described as station $\{F1, F2, F3\}$ and the set shown as $f(c2)$. Both sections $c1$ and $c2$ can be further expanded into sets of new energy stations.

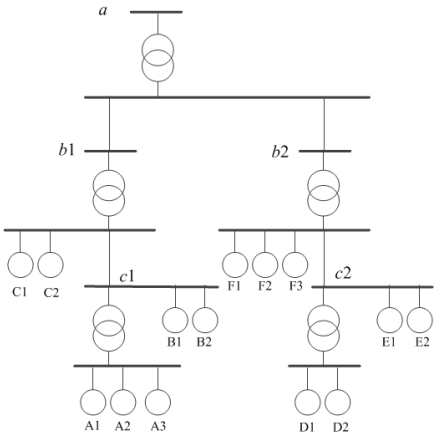


Fig. 3. Schematic diagram of P-V curve for renewable energy power transmission system

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B. The Simulation results of Static Voltage Stability Limit

The complex section coupling relationship makes the power transfer capability of the section highly affected by the regional active power flow distribution. Take the static voltage stability limit of a secondary section's power sending area as an example. The limit of Section A refers to the renewable energy power transfer limit connected to the 500kV substation, while the limit of Section B is the static voltage stability limit of the 220kV gathering station's outgoing lines. Section B is connected to Section A via 220kV transmission lines, and there are also renewable energy stations directly connected to Section A through 220kV lines.

Simulation results show that the calculation results of section power transfer limits vary significantly under different power flow conditions. As shown in Fig.4, when all renewable energy stations in the area output power uniformly, the power transfer limit of Section A is relatively large, while the limit of downstream Section B is relatively small. Increasing the active power transfer level of Section B can indeed improve its own limit, but the power transfer limit of Section A decreases rapidly, leading to a reduction in the overall regional limit.

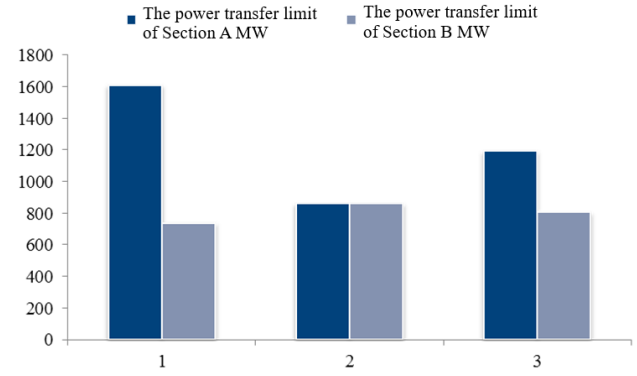


Fig. 4. Static voltage stability limits in a secondary section's power sending area under different power flow conditions (1-Uniform Output, 2-Only Section B has power output; 3- Section B has power output of 800MW, Section A has power output of 1125MW)

IV. Analysis of Static Voltage Stability Limit for Different Section Coupling Types

A. Analysis of Section Coupling Types

By simplifying and extracting typical section coupling types, they can be summarized as parallel coupling transmission systems and series coupling transmission systems, as shown in Fig.5 and Fig.6. In the parallel coupling transmission system of Fig.5, two renewable energy stations are connected in parallel to an infinite bus through their respective transmission lines. In the series coupling transmission system of Fig.6, the remote renewable energy station is connected to the proximal station via a transmission line, and then transmitted to the infinite bus.

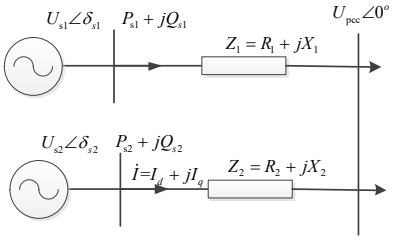


Fig. 5. Parallel coupling transmission system for renewable energy

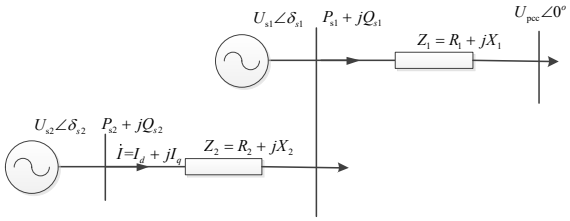


Fig. 6. Series coupling transmission system for renewable energy

B. Parallel Coupling Transmission System

For the parallel coupling transmission system shown in Fig.5, assume that the renewable energy installed capacities of the two branches are identical, and the reactive power input levels are consistent. When $X_1 > X_2$, there is $P_{CC1} < P_{CC2}$. Assume $P_{CC1} = k_1 P_{CC2} (k_1 < 1)$.

If the active power output of the branch corresponding to X_1 is k times that of the branch corresponding to X_2, the overall static voltage stability limit is:

$$P_{\lim} = \begin{cases} (1+k)P_{CC2}, & k \le k_1 \\ (1+\dfrac{1}{k})P_{CC1}, & k > k_1 \end{cases} \tag{2}$$

when $k = k_1$ is met, the overall static stability limit reaches the maximum value $(1+1/k_1)P_{CC1}$. It can be seen that the active power distribution of different parallel branches has a great influence on the static voltage stability limit. Assuming that the active power of renewable energy in different branches increases uniformly, with the increase of the proportion of active power output of the branch with larger transmission reactance, the overall static stability limit of the system shows a process of first increasing and then decreasing, as shown in Fig.7. When the static stability limit takes the maximum value, the proportion of active power output of the branch with larger transmission reactance is less than that of the branch with smaller transmission reactance.

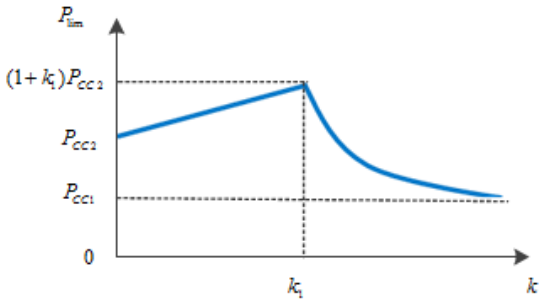


Fig. 7. Influence of renewable energy distribution on static voltage stability in parallel system

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For the parallel coupling transmission system, considering a relatively severe operating condition where all future active power growth occurs in the heavily loaded branch, the current transmission limit is the sum of the transmission limit of the heavily loaded branch and the current transmission power of other branches:

$$\mathrm{d}P_{s2} = P_{cc2} - P_{s2} \tag{3}$$

$$\mathrm{d}P_{s1} = P_{cc1} - P_{s1} \tag{4}$$

$$\mathrm{d}P_{s1} < \mathrm{d}P_{s2} \tag{5}$$

$$P_{\lim} = P_{cc1} + P_{s2} \tag{6}$$

Based on the above current operating conditions, possible future output changes over time can be divided into the following three scenarios:

a. Output Change Scenario 1

$$P_{s1}' = P_{s1} + \Delta P \tag{7}$$

If the output growth comes entirely from the heavily loaded branch, the system transmission limit remains unchanged:

$$P_{\lim}' = P_{cc1} + P_{s2} \tag{8}$$

b. Output Change Scenario 2

$$P_{s2}' = P_{s2} + \Delta P \tag{9}$$

If the output growth comes from the lightly loaded branch, and also:

$$\mathrm{d}P_{s1} < \mathrm{d}P_{s2} - \Delta P \tag{10}$$

Then the system transmission limit is:

$$P_{\lim}' = P_{cc1} + P_{s2} + \Delta P \tag{11}$$

That is, the transmission limit increases by ΔP .

c. Output Change Scenario 3

The output growth comes from the lightly loaded branch, and also:

$$\mathrm{d}P_{s1} \geq \mathrm{d}P_{s2} - \Delta P \tag{12}$$

Then the system transmission limit is:

$$P_{\lim}' = P_{cc2} + P_{s1} \tag{13}$$

It follows that:

$$P_{\lim}' - P_{\lim} = P_{cc2} + P_{s1} = \mathrm{d}p_{s2} - \mathrm{d}p_{s1} > 0 \tag{14}$$

That is, the limit increases by $\mathrm{d}p_{s2} - \mathrm{d}p_{s1}$.

C. Series Coupling Transmission System

For the series coupling transmission system, considering a relatively severe operating condition where all future active power growth occurs at the remote station, the current transmission limit is the sum of the transmission limit of the remote station and the unchanged power output of the proximal station, which is:

$$P_{\lim} = P_{s1} + P_{cc2}\left|P_{s1}\right. \tag{15}$$

Based on the above current operating conditions, possible future output changes over time can be divided into the following two scenarios:

a. Output Change Scenario 1

$$P_{s2}' = P_{s2} + \Delta P \tag{16}$$

If the output growth comes entirely from the remote station, the system transmission limit remains unchanged:

$$P_{\lim}' = P_{s1} + P_{cc2}\left|P_{s1}\right. \tag{17}$$

b. Output Change Scenario 2

$$P_{s1}' = P_{s1} + \Delta P \tag{18}$$

If the output growth comes from the proximal station, the system transmission limit is:

$$P_{\lim}' = P_{s1} + \Delta P + P_{cc2}\left|(P_{s1} + \Delta P)\right. \tag{19}$$

where $P_{cc2}\left|(P_{s1} + \Delta P)\right.$ represents the transmission limit of the remote station when the output of the proximal station is $P_{s1} + \Delta P$. This is a non-linear implicit equation that is difficult to express analytically. For practical systems, the transmission limits under different operating conditions can be obtained through simulation. For online applications, a look-up table method is recommended.

The sensitivity of section transmission limit to key node power flow changes can be obtained through simulation, and the sensitivity calculation formula is:

$$S_{P_{s1},1} = \frac{\Delta P_{\lim}}{\Delta P_{s1}} \tag{20}$$

Fig.8 shows the sensitivity simulation results of the power changes in the proximal station of an actual secondary section's power sending area in Jibei power grids to the section transmission limit. It can be seen that as the power of the proximal station gradually increases, its sensitivity to the section transmission limit shows a gradual decreasing trend. When the active power of the proximal station is close to zero, its sensitivity to the total renewable energy transmission limit of the region is close to 1, meaning that its active power growth hardly causes the transmission limit of the remote section to decrease, and the active power growth of the proximal station can significantly improve the total transmission capacity. When the active power level of the proximal station is below 900 MW, with the increase of its active power, its sensitivity to the total transmission limit decreases slowly. That is, the same active power growth of the proximal station leads to a larger decrease in the transmission limit of the remote section. When the active power level of the proximal station exceeds 900 MW, its sensitivity to the total transmission limit decreases rapidly. At this time, increasing the active power of the proximal station will significantly reduce the transmission limit of the remote section, and its positive contribution to the total transmission limit becomes smaller and smaller.

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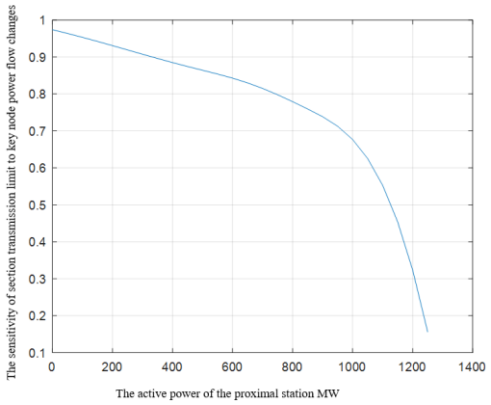


Fig. 8. Sensitivity of section transmission limit to key node power flow

V. Conclusion

This paper studies the static voltage stability issues of renewable energy through multi-voltage-level section transmission systems faced by actual power grids. The complex section coupling problem is decomposed into a series coupling transmission system and a parallel coupling transmission system. A complementary approach combining simplified model analytical analysis and actual system simulation results is adopted to investigate the interactive influence laws of power limits of each coupling section under static voltage stability constraints. The study shows that the power flow distribution of different-level sections and different power flow growth patterns between sections have obvious regular influences on the overall renewable energy transmission capacity of the region. The research findings can provide a theoretical basis for the online assessment of transmission limits in multi-voltage-level section transmission systems. On the premise of ensuring safety, from the perspective of fully releasing the transmission capacity of sections, it is conducive to promoting renewable energy consumption.

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References

[1] S Bai, J Zhang, Y Bao et al., “On-line Limit Power Calculation of Correlative Transmission Section Considering New Energy Consumption,” Power System Technology, vol. 44, pp. 3717–3724, October 2020(in Chinese).

[2] J Wang, Y Wang, Z Yang et al., “An Active Power Control Strategy for Large-scale Cluster of Wind Power Considering Constraints on Nested Transmission Sections,” Automation of Electric Power Systems, vol 39, pp. 16–21, July 2015(in Chinese).

[3] Y Shu, Z Zhang, J Guo et al., “Study on key factors and solution of renewable energy accommodation,” Proceedings of the CSEE, vol. 37, pp. 1–8, Jan 2017(in Chinese).

[4] B Zhao, J Ren, J Chen et al., “Tri-level robust planning-operation co-optimization of distributed energy storage in distribution networks with high PV penetration,” Applied Energy, vol. 279, pp. 115768, December 2020.

[5] P Li. “Analysis of static voltage stability control strategy for power systems with high proportion of wind power integration,” Mechanical and Electrical Information, vol.15, pp. 15–18, 2023(in Chinese).

[6] F Gao, Y Wang, L Guo et al., “A novel load-nodal voltage stability index of power system,” RPG 2015, Beijing, 2015.

[7] S Lian, H Lin. “Research on static voltage stability of power system containing wind power farm,” Modern Electronics Technique, vol.45, pp. 182–186, 2022(in Chinese).

[8] L Wu, W Zhao, M Xu et al., “Mechanism Analysis and Suppression of Repeated Voltage Fluctuation Considering Fault Ride Through Characteristics of the Wind Turbine”, Journal of Global Energy Interconnection, vol. 5, pp. 290-297, May 2022(in Chinese).

[9] J Zhuo, X Jin, B Deng et al., “An Active Power Control Method of Large-Scale Wind Farm Considering Security Constraints of Tie Lines” Power System Technology, vol. 39, pp. 1014-1018, April 2015(in Chinese).

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